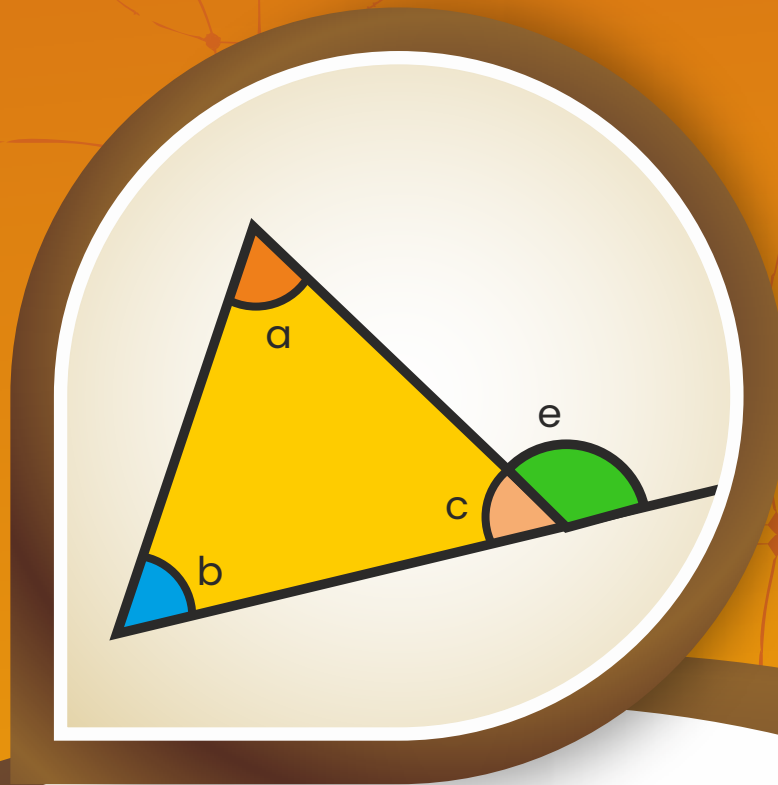


MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Height and Distance

(English Medium)

CHECK YOUR GRASP

HEIGHTS AND DISTANCE

EXERCISE-I

1. The angle of elevation of the top of a tower from a point 20 metre away from its base is 45° . Then height of the tower is-

(1) 10 m (2) 20 m (3) 40 m (4) $20\sqrt{3}$ m

HD0001

2. At a point 15 metre away from the base of a 15 metre high house, the angle of elevation of the top is-

(1) 45° (2) 30° (3) 60° (4) 90°

HD0002

3. An aeroplane flying at a height 300 metre above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. Then the height of the lower plane from the ground in metres is-

(1) $100\sqrt{3}$ (2) $100/\sqrt{3}$
(3) 50 (4) $150(\sqrt{3} + 1)$

HD0003

4. If the elevation of the sun is 30° then the length of the shadow cast by a tower of 150 metres height is-

(1) $75\sqrt{3}$ m (2) $200\sqrt{3}$ m
(3) $150\sqrt{3}$ m (4) None of these

HD0004

5. From the top of the cliff 300 metres high, the top of a tower was observed at an angle of depression 30° and from the foot of the tower the top of the cliff was observed at an angle of elevation 45° . The height of the tower is -

(1) $50(3 - \sqrt{3})$ m (2) $200(3 - \sqrt{3})$ m
(3) $100(3 - \sqrt{3})$ m (4) None of these

HD0005

6. The angles of elevation of the top of a tower at the top and the foot of a pole of height 10 m are 30° and 60° respectively. The height of the tower is-

(1) 10 m (2) 15 m
(3) 20 m (4) None of these

HD0006

7. A person, standing on the bank of a river, observes that the angle subtended by a tree on the opposite bank is 60° ; when he retreats 20 m from the bank, he finds the angle to be 30° . The height of the tree and the breadth of the river are -

(1) $10\sqrt{3}$ m, 10 m (2) 10 m ; $10\sqrt{3}$ m
(3) 20 m, 30m (4) None of these

HD0007

8. From the top of a light house 60 m high with its base at sea level the angle of depression of a boat is 15° . The distance of the boat from the light house is-

(1) $60\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$ m (2) $60\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ m

(3) $30\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$ m (4) $30\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ m

HD0008

9. The angle of elevation of the sun when the length of the shadow of a pole is $\sqrt{3}$ times the height of the pole will be-

(1) 30° (2) 60° (3) 90° (4) 45°

HD0009

10. If the length of the shadow of a vertical pole on the horizontal ground is equal to its height, find the angle of elevation of the sun -

(1) 60° (2) 30° (3) 45° (4) 90°

HD0010

11. The angle of elevation of the top of a tower from a point 20m away from its base is 60° . The height of the tower is -

(1) 10 m (2) $20/\sqrt{3}$ m (3) 40 m (4) $20\sqrt{3}$ m

HD0011

12. A flag staff on the top of the tower 80 meter high, subtends an angle $\tan^{-1}\left(\frac{1}{9}\right)$ at a point on the ground 100 meters away from the foot of the tower. Find the height of the flag-staff -

(1) 20 m (2) 30 m (3) 25 m (4) 35 m

HD0012

23. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar is-

- (1) $\sqrt{3} : 1$ (2) $1 : 3$
(3) $1 : \sqrt{3}$ (4) $\sqrt{3} : 2$

HD0023

24. The shadow of a tower of height $(1 + \sqrt{3})$ metre standing on the ground is found to be 2 metre longer when the sun's elevation is 30° , than when the sun's elevation was -

- (1) 30° (2) 45° (3) 60° (4) 75°

HD0024

25. The angle of elevation of the top of an incomplete vertical pillar at a horizontal distance of 50 m from its base is 45° . If the angle of elevation of the top of the complete pillar the same point is to be 60° , then the height of the incomplete pillar is to be increased by-

- (1) $50(\sqrt{3} - 1)$ m (2) $50(\sqrt{3} + 1)$ m
(3) 50 m (4) $25\sqrt{2}$ m.

HD0025

26. A vertical tower stands on a declivity which is inclined at 15° to the horizon. From the foot of the tower a man ascends the declivity for 80 feet and then finds that tower subtends an angle of 30° . The height of the tower is-

- (1) $20(\sqrt{6} - \sqrt{2})$
(2) $40(\sqrt{6} - \sqrt{2})$
(3) $40(\sqrt{6} + \sqrt{2})$
(4) None of these

HD0026

27. AB is a vertical pole. The point A of pole AB is on the level ground. C is the middle point of AB. P is a point on the level ground. The portion BC subtends an angle β at P. If $AP = n AB$, then $\tan \beta$ is equal to-

- (1) $\frac{n}{2n^2 + 1}$ (2) $\frac{n}{n^2 - 1}$
(3) $\frac{n}{n^2 + 1}$ (4) None of these

HD0027

28. The top of a hill observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. The height of the hill is -

- (1) $\frac{h \cot q}{\cot q - \cot p}$ (2) $\frac{h \cot p}{\cot p - \cot q}$
(3) $\frac{h \tan p}{\tan p - \tan q}$ (4) None of these

HD0028

29. A and B are two points 30 m apart in a line on the horizontal plane through the foot of a tower lying on opposite sides of the tower. If the distance of the top of the tower from A and B are 20 m and 15 m respectively, the angle of elevation of the top of the tower at A is-

- (1) $\cos^{-1}(43/48)$ (2) $\sin^{-1}(43/48)$
(3) $\cos^{-1}(29/36)$ (4) $\sin^{-1}(29/36)$

HD0029

30. A vertical pole subtends an angle $\tan^{-1}(1/2)$ at a point P on the ground. The angle subtended by the upper half of the pole at the point P is-

- (1) $\tan^{-1}(1/4)$ (2) $\tan^{-1}(2/9)$
(3) $\tan^{-1}(1/8)$ (4) $\tan^{-1}(2/3)$

HD0030

31. The angle of elevation of the top of a tower standing on a horizontal plane from a point A is α . After walking a distance d towards the foot of the tower, the angle of elevation is found to be β . The height of the tower is-

- (1) $\frac{d \sin \alpha \sin \beta}{\sin(\beta - \alpha)}$ (2) $\frac{d \sin \alpha \sin \beta}{\sin(\alpha - \beta)}$
(3) $\frac{d \sin(\beta - \alpha)}{\sin \alpha \sin \beta}$ (4) $\frac{d \sin(\alpha - \beta)}{\sin \alpha \sin \beta}$

HD0031

- 32.** The angle of elevation of the top of two vertical towers as seen from the middle point of the line joining the foot of the towers are 60° and 30° respectively. The ratio of the height of the towers is-

- (1) $2:1$ (2) $\sqrt{3}:1$
(3) $3:2$ (4) $3:1$

HD0032

- 33.** A person walking along a st. road towards a hill observes at two points, distance $\sqrt{3}$ km, the angles of elevation of the hill to be 30° and 60° . The height of the hill is-

- $$\begin{array}{ll} (1) \frac{3}{2} \text{ km} & (2) \sqrt{\frac{2}{3}} \text{ km} \\ (3) \frac{\sqrt{3}+1}{2} \text{ km} & (4) \sqrt{3} \text{ km} \end{array}$$

HD0033

- 34.** The length of the shadow of a vertical pole of height h , thrown by the sun's rays at three different moments are h , $2h$ and $3h$. The sum of the angles of elevation of the rays at these three moments is equal to-

-
- (1) $\frac{\pi}{2}$
- (2) $\frac{\pi}{3}$
- (3) $\frac{\pi}{4}$
- (4) $\frac{\pi}{6}$

HD0034

- 35.** The upper $\frac{3}{4}$ th portion of a vertical pole subtends an angle $\tan^{-1} \frac{3}{5}$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is-

[AIEEE-2002]

- (1) 80 m (2) 20 m
(3) 40 m (4) 60 m

HD0035

- 36.** A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30° . The breadth of the river is-

[AIEEE-2004]

- (1) 20 m (2) 30 m
(3) 40 m (4) 60 m

HD0036

- 37.** A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that $AB (=a)$ subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30° . The height of the tower is- **[AIEEE-2007]**

[AIEEE-2007]

- (1) $2a/\sqrt{3}$ (2) $2a\sqrt{3}$
(3) $a/\sqrt{3}$ (4) $a\sqrt{3}$

HD0037

- 38.** AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60° . He moves away from the pole along the line BC to a point D such that $CD = 7$ m. From D the angle of elevation of the point A is 45° . Then the height of the pole is-
- [AIEEE-2008]**

[AIEEE-2008]

- (1) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3}-1} m$
- (2) $\frac{7\sqrt{3}}{2} (\sqrt{3}+1) m$
- (3) $\frac{7\sqrt{3}}{2} (\sqrt{3}-1) m$
- (4) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3}+1} m$

HD0038

- 39.** A man standing on a horizontal plane, observes the angle of elevation of the top of a tower to be α . After walking a distance equal to double the height of the tower, the angle of elevation becomes 2α , then α is -

- (1) $\frac{\pi}{18}$

(3) $\frac{\pi}{6}$

(2) $\frac{\pi}{12}$

(4) $\frac{\pi}{2}$

HD0039

40. ABCD is a trapezium such that AB and CD are parallel and $BC \perp CD$. If $\angle ADB = \theta$, $BC = p$ and $CD = q$, then AB is equal to [JEE-MAINS-2013]

$$(1) \frac{(p^2 + q^2)\sin \theta}{p \cos \theta + q \sin \theta} \quad (2) \frac{p^2 + q^2 \cos \theta}{p \cos \theta + q \sin \theta}$$

$$(3) \frac{p^2 + q^2}{p^2 \cos \theta + q^2 \sin \theta} \quad (4) \frac{(p^2 + q^2)\sin \theta}{(p \cos \theta + q \sin \theta)^2}$$

HD0040

41. If the angles of elevation of the top of a tower from three collinear points A, B and C, on a line leading to the foot of the tower, are 30° , 45° and 60° respectively, then the ratio, AB : BC, is :

[JEE(Main)-2015]

$$(1) 1 : \sqrt{3} \quad (2) 2 : 3$$

$$(3) \sqrt{3} : 1 \quad (4) \sqrt{3} : \sqrt{2}$$

HD0041

42. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30° . After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60° . Then the time taken (in minutes) by him, from B to reach the pillar, is :

[JEE(Main)-2016]

$$(1) 5 \quad (2) 6 \quad (3) 10 \quad (4) 20$$

HD0042

43. Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that $AP = 2AB$. If $\angle BPC = \beta$, then $\tan \beta$ is equal to :- [JEE(Main)-2017]

$$(1) \frac{4}{9} \quad (2) \frac{6}{7} \quad (3) \frac{1}{4} \quad (4) \frac{2}{9}$$

HD0043

44. PQR is a triangular park with $PQ = PR = 200$ m. A T.V. tower stands at the mid-point of QR. If the angles of elevation of the top of the tower at P, Q and R are respectively 45° , 30° and 30° , then the height of the tower (in m) is- [JEE(Main)-2018]

$$(1) 50 \quad (2) 100\sqrt{3}$$

$$(3) 50\sqrt{2} \quad (4) 100$$

HD0044

45. Consider a triangular plot ABC with sides $AB=7$ m, $BC=5$ m and $CA=6$ m. A vertical lamp-post at the mid point D of AC subtends an angle 30° at B. The height (in m) of the lamp-post is :

[JEE(Main) 2019]

$$(1) 7\sqrt{3} \quad (2) \frac{2}{3}\sqrt{21}$$

$$(3) \frac{3}{2}\sqrt{21} \quad (4) 2\sqrt{21}$$

HD0045

46. If the angle of elevation of a cloud from a point P which is 25 m above a lake be 30° and the angle of depression of reflection of the cloud in the lake from P be 60° , then the height of the cloud (in meters) from the surface of the lake is :

[JEE(Main) 2019]

$$(1) 42 \quad (2) 50 \quad (3) 45 \quad (4) 60$$

HD0046

47. Two vertical poles of heights, 20m and 80m stand a part on a horizontal plane. The height (in meters) of the point of intersection of the lines joining the top of each pole to the foot of the other, from this horizontal plane is :

[JEE(Main) Apr-19]

$$(1) 12 \quad (2) 15 \quad (3) 16 \quad (4) 18$$

HD0047

48. Two poles standing on a horizontal ground are of heights 5m and 10 m respectively. The line joining their tops makes an angle of 15° with ground. Then the distance (in m) between the poles, is :-

[JEE(Main) 2019]

$$(1) \frac{5}{2}(2 + \sqrt{3}) \quad (2) 5(\sqrt{3} + 1)$$

$$(3) 5(2 + \sqrt{3}) \quad (4) 10(\sqrt{3} - 1)$$

HD0048

49. ABC is a triangular park with $AB = AC = 100$ metres. A vertical tower is situated at the mid-point of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}(3\sqrt{2})$ and $\operatorname{cosec}^{-1}(2\sqrt{2})$ respectively, then the height of the tower (in metres) is :

[JEE(Main) 2019]

$$(1) 10\sqrt{5} \quad (2) \frac{100}{3\sqrt{3}}$$

$$(3) 20 \quad (4) 25$$

HD0049

50. A 2 m ladder leans against a vertical wall. If the top of the ladder begins to slide down the wall at the rate 25 cm/sec., then the rate (in cm/sec.) at which the bottom of the ladder slides away from the wall on the horizontal ground when the top of the ladder is 1 m above the ground is : [JEE(Main) 2019]

- (1) $25\sqrt{3}$ (2) 25
(3) $\frac{25}{\sqrt{3}}$ (4) $\frac{25}{3}$

HD0050

51. The angle of elevation of the top of vertical tower standing on a horizontal plane is observed to be 45° from a point A on the plane. Let B be the point 30 m vertically above the point A. If the angle of elevation of the top of the tower from B be 30° , then the distance (in m) of the foot of the tower from the point A is: [JEE(Main) 2019]

- (1) $15(3 - \sqrt{3})$ (2) $15(3 + \sqrt{3})$
(3) $15(1 + \sqrt{3})$ (4) $15(5 - \sqrt{3})$

HD0051

52. Two vertical poles AB = 15 m and CD = 10 m are standing apart on a horizontal ground with points A and C on the ground. If P is the point of intersection of BC and AD, then the height of P (in m) above the line AC is : [JEE(Main) 2020]

- (1) $20/3$ (2) 5
(3) $10/3$ (4) 6

HD0052

53. The angle of elevation of a cloud C from a point P, 200 m above a still lake is 30° . If the angle of depression of the image of C in the lake from the point P is 60° , then PC (in m) is equal to : [JEE(Main) 2020]

- (1) 400 (2) $400\sqrt{3}$
(3) 100 (4) $200\sqrt{3}$

HD0053

54. The angle of elevation of the top of a hill from a point on the horizontal plane passing through the foot of the hill is found to be 45° . After walking a distance of 80 meters towards the top, up a slope inclined at an angle of 30° to the horizontal plane, the angle of elevation of the top of the hill becomes 75° . Then the height of the hill (in meters) is_. [JEE(Main) 2020]

HD0054

55. The angle of elevation of the summit of a mountain from a point on the ground is 45° . After climbing up one km towards the summit at an inclination of 30° from the ground, the angle of elevation of the summit is found to be 60° . Then the height (in km) of the summit from the ground is : [JEE(Main) 2020]

- (1) $\frac{1}{\sqrt{3}-1}$ (2) $\frac{1}{\sqrt{3}+1}$
(3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

HD0055

56. A spherical gas balloon of radius 16 meter subtends an angle 60° at the eye of the observer A while the angle of elevation of its center from the eye of A is 75° . Then the height (in meter) of the top most point of the balloon from the level of the observer's eye is : [JEE(Main) 2021]

HD0056

- (1) $8(2 + 2\sqrt{3} + \sqrt{2})$ (2) $8(\sqrt{6} + \sqrt{2} + 2)$
(3) $8(\sqrt{2} + 2 + \sqrt{3})$ (4) $8(\sqrt{6} - \sqrt{2} + 2)$

57. Two poles, AB of length a metres and CD of length a + b (b ≠ a) metres are erected at the same horizontal level with bases at B and D. If BD = x

and $\tan \angle ACB = \frac{1}{2}$, then: [JEE(Main) 2021]

- (1) $x^2 + 2(a + 2b)x - b(a + b) = 0$
(2) $x^2 + 2(a + 2b)x + a(a + b) = 0$
(3) $x^2 - 2ax + b(a + b) = 0$
(4) $x^2 - 2ax + a(a + b) = 0$

HD0057

58. A man is observing, from the top of a tower, a boat speeding towards the tower from a certain point A, with uniform speed. At that point, angle of depression of the boat with the man's eye is 30° (Ignore man's height). After sailing for 20 seconds, towards the base of the tower (which is at the level of water), the boat has reached a point B, where the angle of depression is 45° . Then the time taken (in seconds) by the boat from B to reach the base of the tower is : **[JEE(Main) 2021]**

- (1) 10 (2) $10\sqrt{3}$
(3) $10(\sqrt{3}+1)$ (4) $10(\sqrt{3}-1)$

HD0058

59. Two vertical poles are 150 m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is : **[JEE(Main) 2021]**

- (1) $20\sqrt{3}$ (2) $25\sqrt{3}$ (3) 30 (4) 25

HD0059

60. Let AB and PQ be two vertical poles, 160 m apart from each other. Let C be the middle point of B

and Q, which are feet of these two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A, respectively. If the height of pole PQ is twice the height of pole AB, then $\tan^2\theta$ is equal to

[JEE(Main) 2022]

- (A) $\frac{3-2\sqrt{2}}{2}$ (B) $\frac{3+\sqrt{2}}{2}$
(C) $\frac{3-2\sqrt{2}}{4}$ (D) $\frac{3-\sqrt{2}}{4}$

HD0060

61. A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that $QR = 15$ m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is :

[JEE(Main) 2022]
HD0061

- (A) $5(2\sqrt{3}+3)$ m (B) $5(\sqrt{3}+3)$ m
(C) $10(\sqrt{3}+1)$ m (D) $10(2\sqrt{3}+1)$ m

62. The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60° . If $\angle BAQ = 30^\circ$, $AB = d$ and the area of the trapezium PQRB is α , then the ordered pair (d, α) is :

[JEE(Main) 2022]

- (A) $(10(\sqrt{3}-1), 25)$ (B) $(10(\sqrt{3}-1), \frac{25}{2})$
(C) $(10(\sqrt{3}+1), 25)$ (D) $(10(\sqrt{3}+1), \frac{25}{2})$

HD0062

63. The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is

$\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot\alpha$ is equal to :

[JEE(Main) 2022]

- (A) $\frac{6}{5}$ (B) $\frac{9}{5}$ (C) $\frac{4}{3}$ (D) $\frac{7}{3}$

HD0063

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	2	1	1	3	3	2	1	2	1	3	4	1	1	1	2
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	1	2	2	2	4	3	3	3	2	1	2	1	2	1	2
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	1	4	1	1	3	1	3	2	2	1	3	1	4	4	2
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	3	3	3	3	2	4	1	80.00	1	2	3	3	2	C
Que.	61	62	63												
Ans.	A	A	A												

Important Notes

